

IB Gateway Setup Guide

SPX 0DTE automated executor

Paper tra

Prepared for Michael | Magis Capital Partners | July 2026

This document explains how Interactive Brokers (IB) Gateway fits into our SPX 0DTE trading system, what you need to configure on your IB account, and how to run a paper session before we go live. The Python strategy (ib_executor.py) does not connect to IB directly - it connects to IB Gateway running on your computer.

1. What is IB Gateway?

IB Gateway is Interactive Brokers' lightweight trading application. It has no charts or full trading screens. Its job is to maintain a logged-in session to IB and expose a socket API that our Python program uses for quotes, orders, and fills.

Think of it as a bridge:

Python executor (our strategy)	--> localhost:7497 (API port)	--> IB Gateway (your login)	--> IB servers (orders + data)
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Gateway vs Trader Workstation (TWS)

- Gateway: headless, low resource - best for automated trading.
- TWS: full UI with charts - best for manual trading.
- Both use the same API ports and rules. We recommend Gateway for the bot.

2. Paper vs live - ports and logins

IB uses separate ports and separate usernames for paper and live:

Mode	Port	Login
Paper (simulated)	7497	Paper username (separate from live)
Live (real money)	7496	Live username

Our executor is currently configured for paper mode (port 7497). Orders are simulated; no real money is at risk. Market data behavior still depends on your subscriptions.

3. Market data - why paper does not auto-inherit live subs

Paying for US Index (SPX) and OPRA (options) on your live account does NOT automatically give your paper account real-time data. By default, paper only gets delayed data unless you explicitly share subscriptions.

One-time setup in IB Client Portal (live login)

- Log in to Client Portal with your LIVE account.
- Click Welcome (top right) -> Settings.
- Under Account Configuration, click Paper Trading Account.
- Set "Share real-time market data subscriptions with paper trading account" to YES.
- In the dropdown, select the LIVE username that owns the Index + OPRA subscriptions.
- Click Save.

Market Data API acknowledgement (required for our Python bot)

- Client Portal -> Settings -> Market Data Subscriptions.
- Find Market Data API Acknowledgement (gear icon) -> enable Yes and sign the form.
- Log out of all TWS/Gateway sessions and log back in after signing.

Important: only one session at a time

When market data is shared between live and paper, IB allows only ONE active session to use that data at a time. If live TWS/Gateway is open, paper may show error 10168 (not subscribed). Close live sessions before running paper Gateway for the bot.

4. Installing and configuring IB Gateway

Install

- Download IB Gateway from interactivebrokers.com (same account as TWS).
- Install on the machine that will run the executor during market hours.

API settings (Configure -> Settings -> API -> Settings)

- Enable ActiveX and Socket Clients: ON (required).
- Socket port: 7497 for paper, 7496 for live.
- Read-Only API: OFF (we place orders).
- Trusted IP addresses: 127.0.0.1 if Python runs on the same PC.
- Allow connections from localhost only unless remote access is needed.

Market data in Gateway

- Settings -> Market Data -> enable Allow delayed market data (fallback if live fails).
- For real-time: subs + paper sharing (Section 3) must be complete.

5. Daily workflow (paper session)

Before 9:30 AM Eastern on a trading day:

- Close any live TWS/Gateway sessions.
- Start IB Gateway -> Paper Trading login -> confirm port 7497.
- Optional check: add SPX to a watchlist - price should update (live, not blank).
- On the same machine, open PowerShell in the project folder:

```
cd C:\Users\...\spx-0dte
python scripts\refresh_live_baselines.py
python live\ib_executor.py
```

After the close:

- Stop the executor (Ctrl+C if still running).
- Quit IB Gateway.

Session logs are saved under data/live/YYYY-MM-DD/ (fills, tranches, IB errors).

6. What success looks like in the console

Good startup (real-time data):

```
IB connected (port 7497, market_data_type=1 requested)
[market] SPX spot OK with market_data_type=1 (live)
[market] streaming subscribed lines=80 ... spot=6xxx.x market_data_type=1 (live)
```

Acceptable fallback (delayed - OK for testing, not for production):

```
IB error 10168 ... [SPX]
[market] falling back to delayed market data (type 3)
[market] using delayed quotes (type 3) ...
```

Error 10168 on the first line is normal when live SPX is unavailable; the bot then falls back to 15-minute

delayed data if enabled in Gateway.

7. Troubleshooting

10168 - Requested market data is not subscribed

- Paper sharing not enabled in Client Portal (Section 3).
- Live TWS/Gateway still open - close it.
- Gateway logged in with wrong user (must be paper user on 7497).
- Subscriptions still Pending - often activate next business day.
- Market Data API Acknowledgement not signed.

Connection refused

- Gateway not running, or wrong port (7497 vs 7496).

No tranche activity

- Not an eligible SPXW day, VIX gate skipped session, or market closed.

8. Roles - who does what

Task	Owner
IB account, market data subscriptions	Michael
Enable paper data sharing + API acknowledgement	Michael (Client Portal)
Start/stop IB Gateway before/after RTH	Michael (or designated machine)
Run refresh_live_baselines + ib_executor	Michael's PC or shared trading box
Strategy code, config, debugging	Drew / dev team

9. Subscriptions needed for SPX 0DTE

- CBOE US Index (SPX spot) - or US Securities Snapshot bundle including indices.
- OPRA (US options) - required for SPXW option chain quotes.
- Account minimum: typically \$500 equity + subscription fees (IBKR Pro).

Verify in Client Portal -> Market Data Subscriptions that both show Active (not Pending).

10. Moving from paper to live (later)

- Gateway login: live user, port 7496.
- live_config.py: mode='live', allow_live=True, account_equity set to real NAV.
- Size contracts_per_tranche for production risk limits.
- Run only after paper sessions reconcile cleanly with the backtest.

Repository: *spx-0dte* | Executor: *live/ib_executor.py* | Config: *live/live_config.py* | Support: Drew Goldman